MAGASA FAIZ | BUSINESS INTELLIGENCE BATCH 18

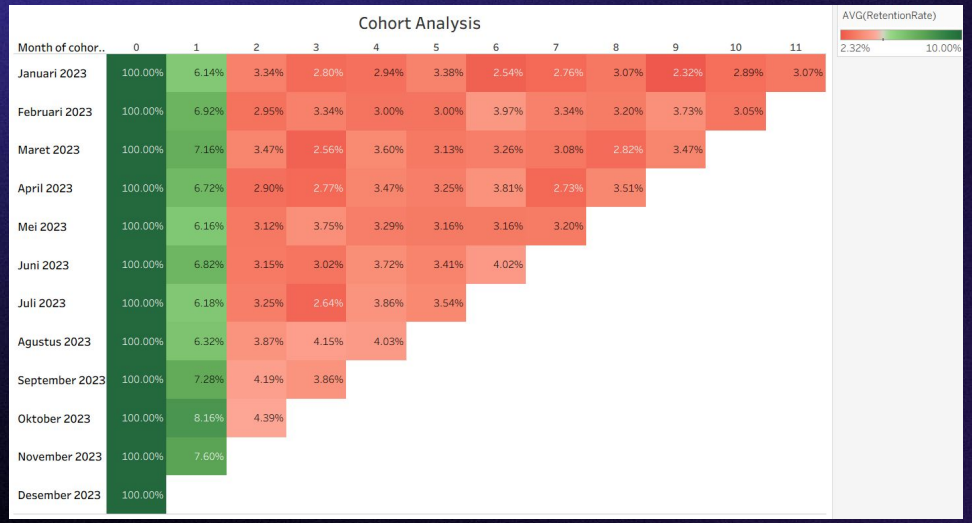
COHORT AND FUNNEL ANALYSIS

Introduction: Purpose of analysis and methods used.

The purpose of this analysis is to understand user retention behavior over time by grouping users based on their first activity month. This cohort analysis helps identify how long users remain active after acquisition and highlights patterns of early churn and long-term retention.

The purpose of this funnel analysis is to understand how users progress through different stages of purchasing behavior, from first-time buyers to repeat and loyal customers. This analysis focuses on identifying conversion drop-offs between purchase stages and evaluating how user volume and profit are distributed across each stage of the funnel. By examining these transitions, the funnel analysis helps highlight where users are most likely to disengage and where retention and revenue optimization strategies can be applied.

User Retention Analysis by Monthly Cohort



Cohort Analysis

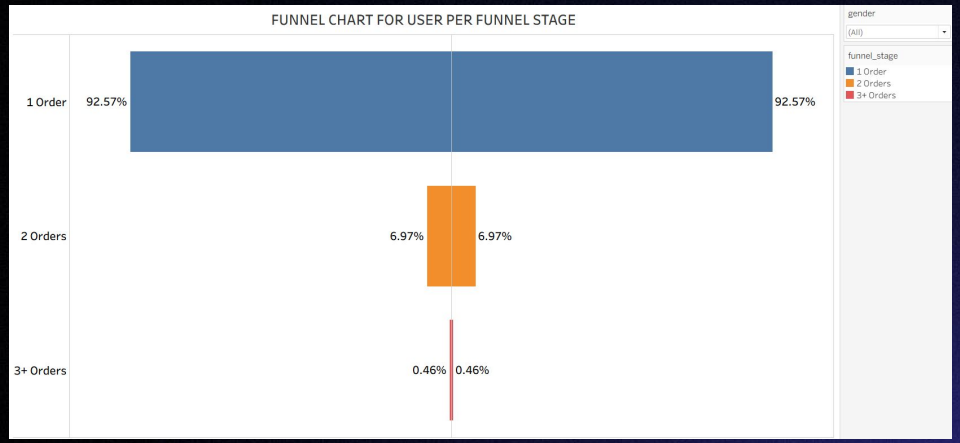
Month of cohor..	0	1	2	3	4	5	6	7	8	9	10	11
Januari 2023	100.00%	6.14%	3.34%	2.80%	2.94%	3.38%	2.54%	2.76%	3.07%	2.32%	2.89%	3.07%
Februari 2023	100.00%	6.92%	2.95%	3.34%	3.00%	3.00%	3.97%	3.34%	3.20%	3.73%	3.05%	
Maret 2023	100.00%	7.16%	3.47%	2.56%	3.60%	3.13%	3.26%	3.08%	2.82%	3.47%		
April 2023	100.00%	6.72%	2.90%	2.77%	3.47%	3.25%	3.81%	2.73%	3.51%			
Mei 2023	100.00%	6.16%	3.12%	3.75%	3.29%	3.16%	3.16%	3.20%				
Juni 2023	100.00%	6.82%	3.15%	3.02%	3.72%	3.41%	4.02%					
Juli 2023	100.00%	6.18%	3.25%	2.64%	3.86%	3.54%						
Agustus 2023	100.00%	6.32%	3.87%	4.15%	4.03%							
September 2023	100.00%	7.28%	4.19%	3.86%								
Oktober 2023	100.00%	8.16%	4.39%									
November 2023	100.00%	7.60%										
Desember 2023	100.00%											

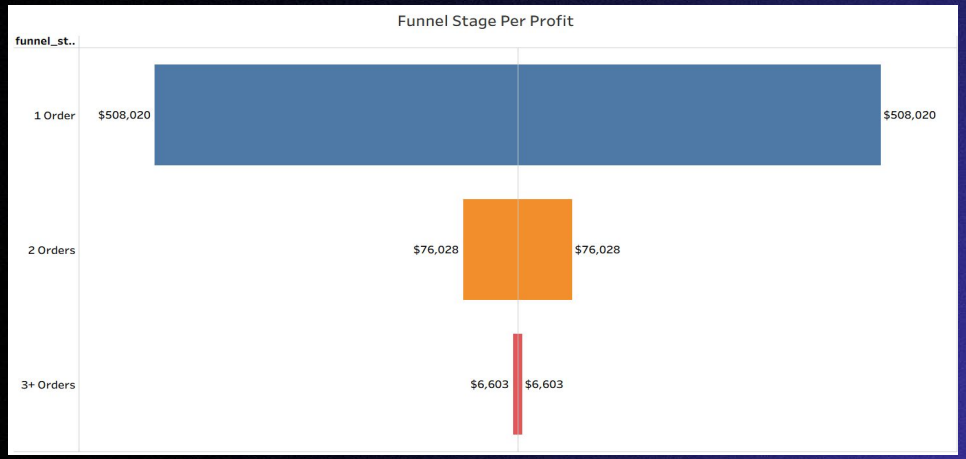
AVG(RetentionRate)
2.32% 10.00%

"The cohort analysis results show a very high churn rate in the first month, with only around 6–8% of users remaining until the second month. However, after the second month, the retention rate tends to stabilize at around 2–4%, indicating the presence of a loyal user base. Furthermore, the cohort at the end of 2023 showed an increase in initial retention compared to the previous cohort, indicating improvements in acquisition quality or user experience. Therefore, the focus of business strategy should be directed at improving onboarding and the initial user experience to increase long-term retention."

From First Purchase to Repeat Buyers: Customer Funnel Analysis

INSIGHT AND RECOMMENDATIONS



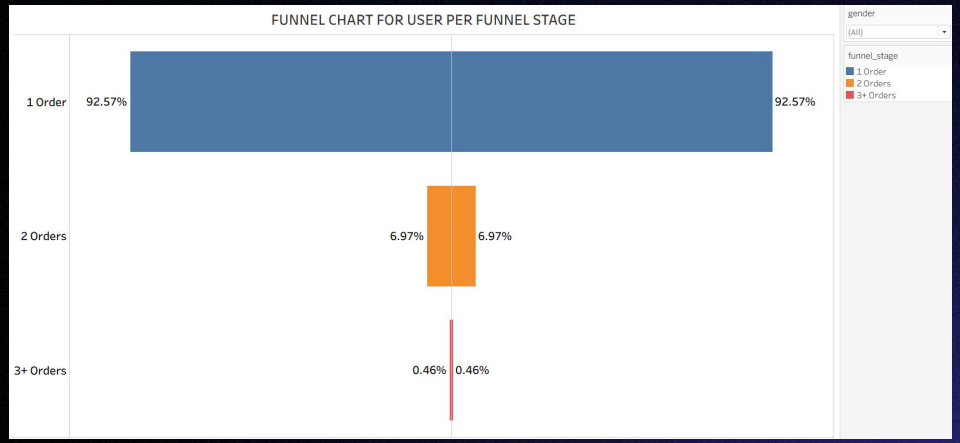


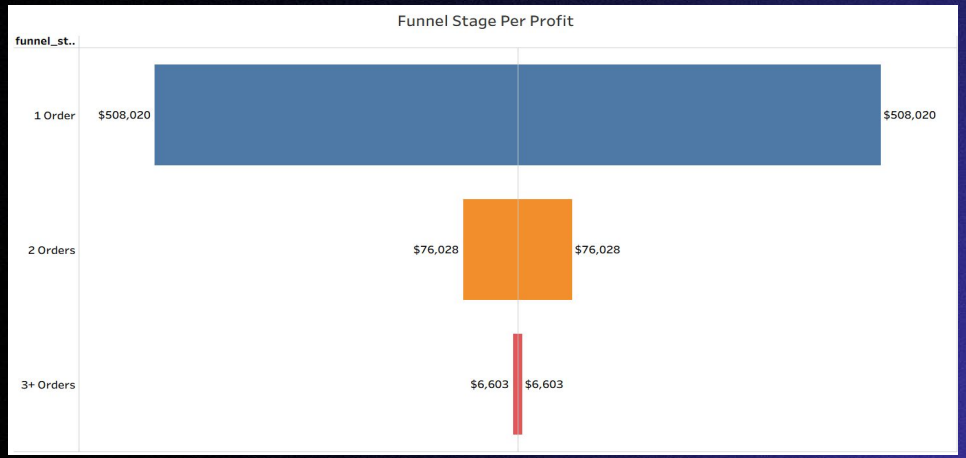
Based on the funnel analysis, the majority of users are concentrated at the first purchase stage. A total of **11,380 users (92.57%)** completed only one order, generating a total profit of **$508,020**.

The number of users drops significantly at the second purchase stage, with only **857 users (6.97%)**, contributing **$76,028** in total profit. The final stage, representing highly loyal customers with **3 or more orders**, consists of only **56 users (0.46%)**, generating **$6,603** in total profit.

This pattern indicates a clear customer lifecycle progression, from first-time buyers to repeat buyers and a small group of loyal customers. The most critical drop-off occurs after the first purchase, suggesting that early-stage retention is a major challenge for the business.

INSIGHT AND RECOMMENDATIONS

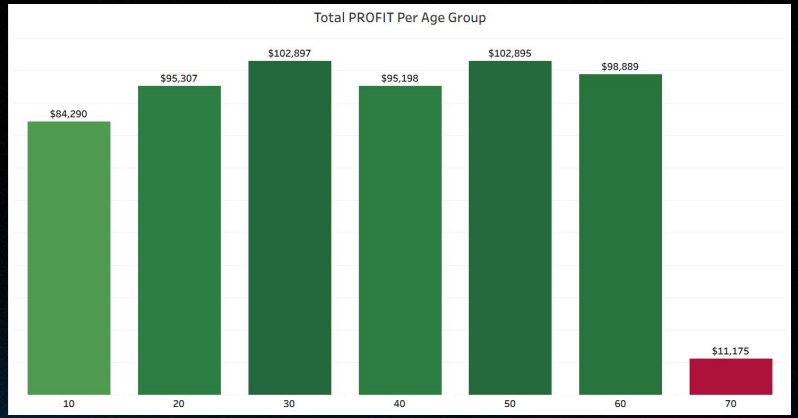


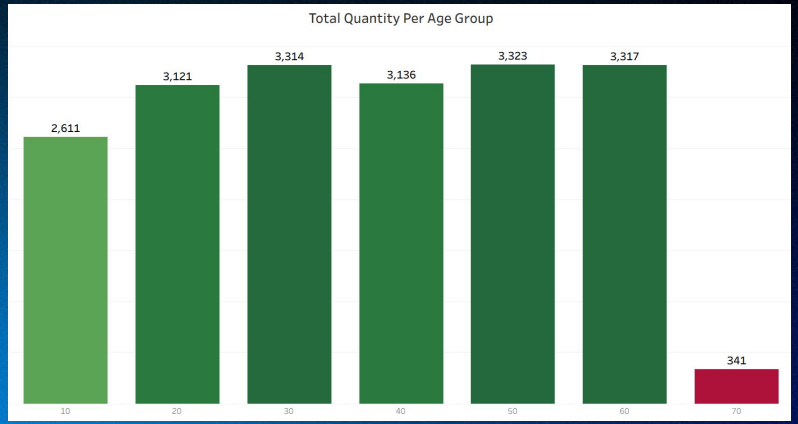


These findings are consistent with the cohort analysis results, which show relatively high initial engagement in the early months, followed by high churn. However, the cohort analysis also indicates the presence of a small but consistent group of loyal customers, with retention rates around **2–4%**, and a noticeable improvement in retention toward the later periods.

Based on these insights, the business may benefit more from prioritizing **customer retention strategies** rather than focusing solely on increasing customer acquisition. Improving the transition from the first to the second purchase should be a key objective, as this stage represents the largest loss of potential long-term customers.

INSIGHT AND RECOMMENDATIONS



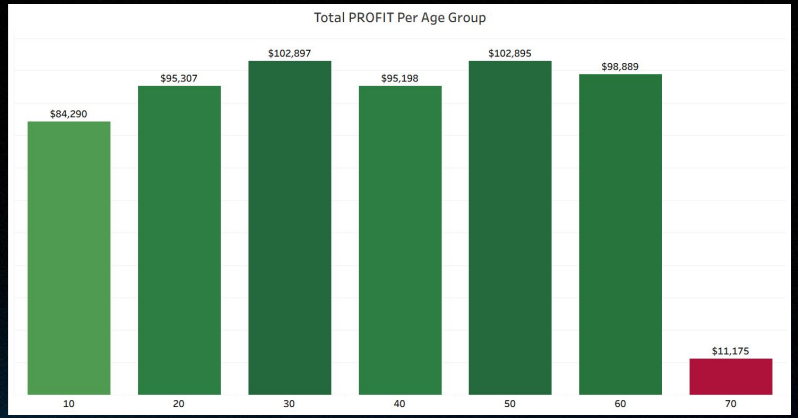


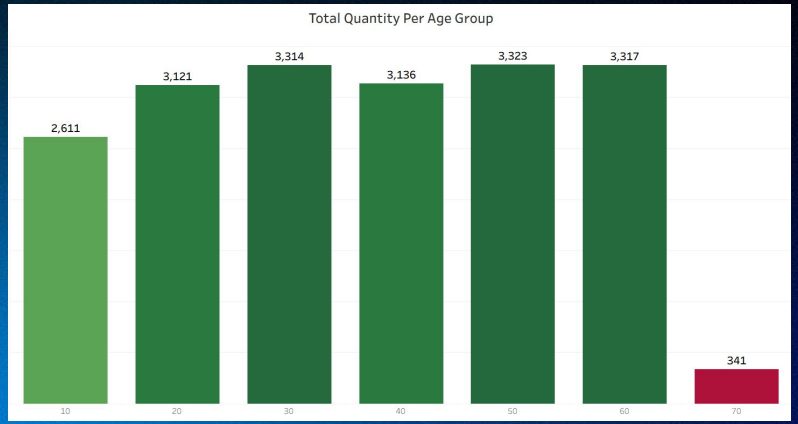
Based on the analysis of **Total Users per Age Group**, **Total Quantity per Age Group**, and **Total Profit per Age Group**, customers were divided into seven age groups using 10-year intervals.

The age group **30–39** shows the highest total profit at **$102,897**, supported by relatively high total quantity (**3,314**) and total users (**2,090**). Similarly, the **50–59** age group also demonstrates strong performance, generating **$102,895** in total profit with the highest total quantity (**3,323**) among all groups.

Overall, the age groups from **10–19 to 60–69** exhibit a relatively stable distribution in terms of users, quantity, and profit contribution. Although the **10–19** age group has lower total users and quantity, its total profit contribution (**$84,890**) is still meaningful and not drastically behind several other age groups.

INSIGHT AND RECOMMENDATIONS



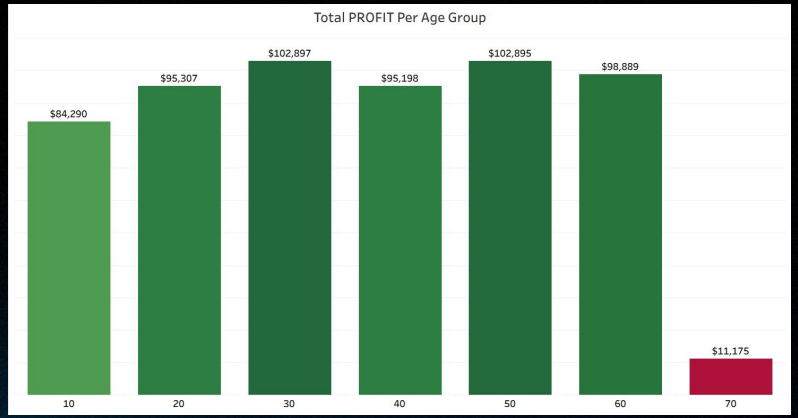


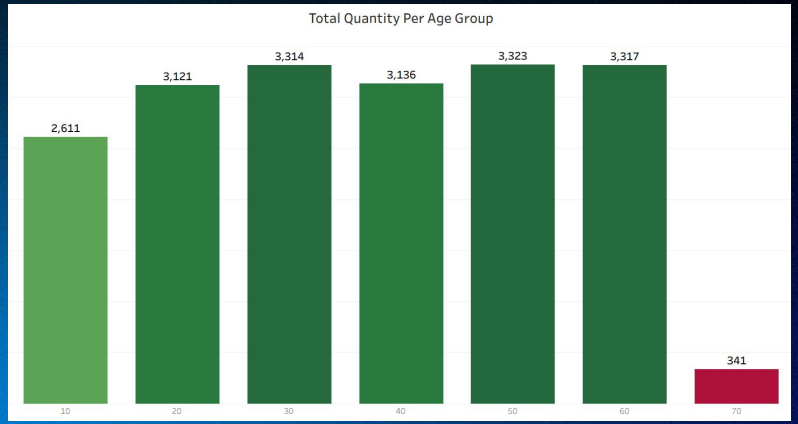
A significant decline is observed in the **70–79** age group, which records substantially lower total users (**205**), quantity (**341**), and profit (**$11,175**). This sharp decrease suggests a clear behavioral or preference shift among older customers.

These patterns indicate that customer purchasing behavior remains relatively consistent across most age groups up to 60 years old, before declining sharply in the oldest segment.

Based on these insights, the business may consider prioritizing marketing strategies toward age groups that demonstrate higher profitability, particularly customers aged **30–39** and **50–59**. These segments show strong purchasing volume and profit contribution, indicating high commercial potential.

INSIGHT AND RECOMMENDATIONS

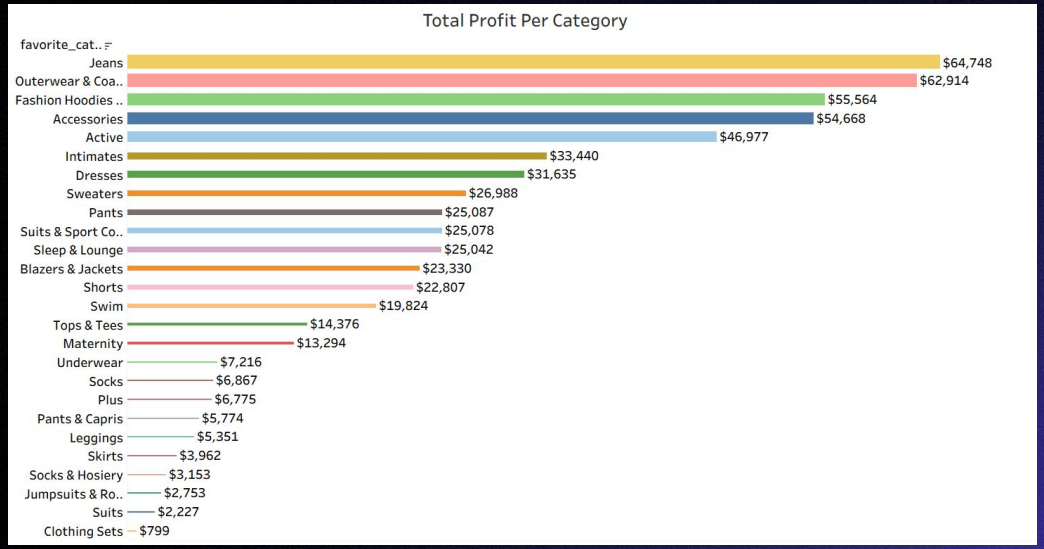




Additionally, the decline in performance among customers aged 70 and above may reflect more focused or limited product preferences. This suggests an opportunity to either design more targeted product offerings for this segment or strategically allocate marketing resources toward age groups with higher growth and profitability potential.

Targeted promotions, age-relevant campaigns, and personalized marketing strategies could help maximize revenue while maintaining a balanced customer portfolio across age groups.

INSIGHT AND RECOMMENDATIONS

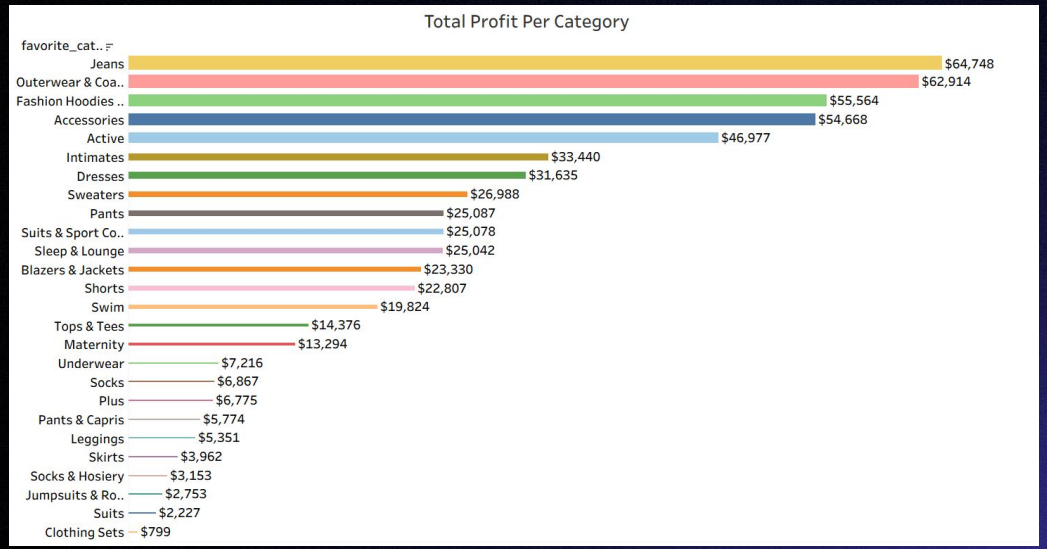


Based on the analysis of **Total Users per Category**, **Total Quantity per Category**, and **Total Profit per Category**, several important patterns can be observed.

Some product categories attract a relatively high number of users, but this does not always translate into the highest quantity purchased or the highest profit generated. Conversely, certain categories generate strong profit despite not having the highest user count, indicating differences in pricing strategy, product value, or profit margins.

This indicates that category performance is not driven by a single metric but by a combination of user engagement, purchasing volume, and product value.

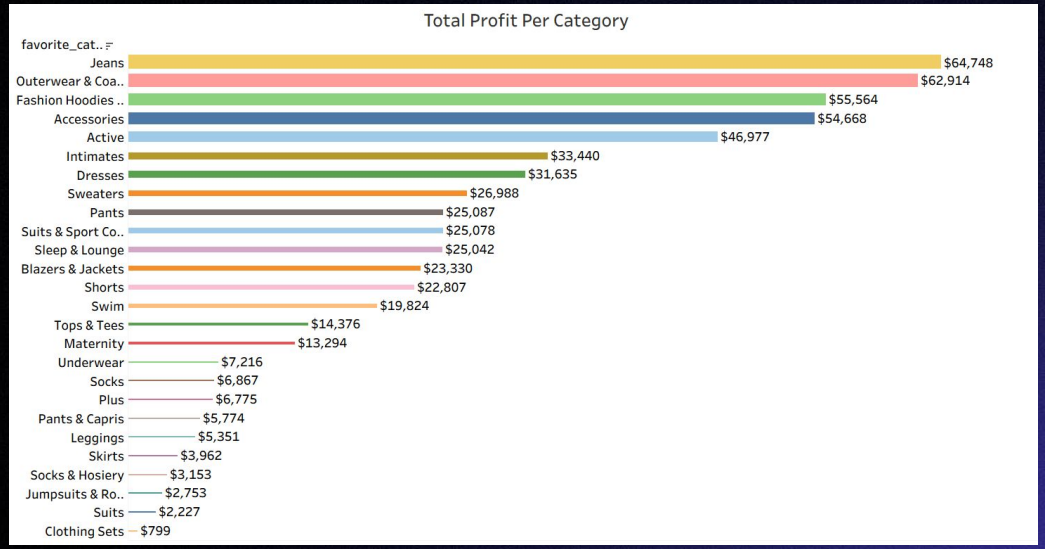
INSIGHT AND RECOMMENDATIONS



Although the ranking of categories varies across the three charts, the shifts are not extreme, suggesting that overall category performance remains relatively consistent while still reflecting different business roles. Broadly, categories can be grouped into three types:

1. Categories that generate high profit mainly due to higher product value rather than volume.
2. Categories that rely more on high purchase quantity to drive revenue and profit.
3. Categories that perform relatively low across users, quantity, and profit.

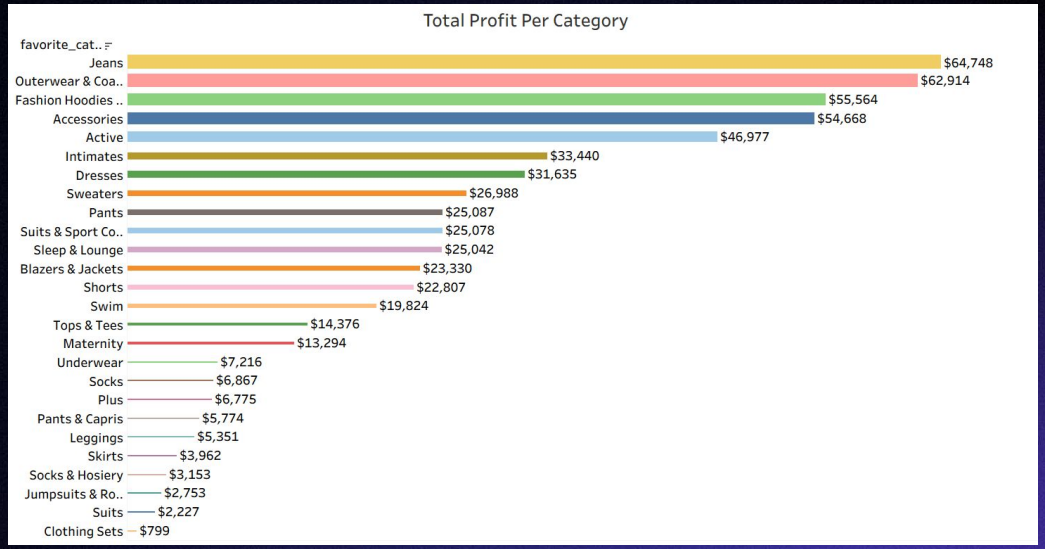
INSIGHT AND RECOMMENDATIONS



Based on these observations, several strategic actions may be considered. Categories with high user engagement but lower profit could benefit from **bundling strategies**, **cross-selling**, or **value enhancement initiatives** to increase average transaction value.

At the same time, categories that generate strong profit despite lower quantity should be maintained and strengthened through **brand positioning**, **product differentiation**, and targeted marketing strategies to preserve their value perception.

INSIGHT AND RECOMMENDATIONS



For categories that perform relatively low across all three metrics, a deeper review may be required to evaluate pricing, product relevance, or promotional effectiveness. Strategic bundling with stronger categories or focused marketing campaigns could help improve their overall contribution. Overall, combining volume-driven and value-driven strategies across categories can help create a more balanced and sustainable revenue structure.

REFLECTIVE QUESTION :

1. Why is data understanding and preprocessing a crucial step before conducting funnel analysis and cohort analysis? Explain the risks that can arise if analysis is performed on data that is not yet clean or whose structure is not yet understood.
2. How do funnel analysis and cohort analysis complement each other in helping businesses understand customer behavior? Explain the different insights generated by each analysis and why each is important.

REFLECTIVE QUESTION FIRST QUESTION

Data understanding and preprocessing are crucial steps before conducting funnel and cohort analysis because both analyses rely heavily on the accuracy and consistency of user-level and time-based data. Understanding the dataset structure ensures that key variables such as user identifiers, timestamps, and transaction stages are correctly interpreted and used in the analysis.

If the data is not properly cleaned or its structure is not well understood, several risks may arise. Duplicate users, missing values, incorrect data types, or inconsistent timestamps can lead to incorrect funnel stage counts or misleading cohort groupings. For example, users may be counted multiple times in the same funnel stage or assigned to the wrong cohort month, which would distort conversion rates and retention metrics.

As a result, insights derived from unclean or misunderstood data may lead to inaccurate conclusions and poor business decisions. Therefore, data understanding and preprocessing help ensure that funnel and cohort analyses reflect actual user behavior rather than data quality issues.

REFLECTIVE QUESTION SECOND QUESTION

Funnel analysis and cohort analysis complement each other by providing different but interconnected perspectives on customer behavior. Funnel analysis focuses on how users progress through different stages of a business process, such as moving from a first purchase to repeat purchases. It helps identify where the largest drop-offs occur and which stages require optimization to improve conversion and revenue.

In contrast, cohort analysis focuses on user retention over time by grouping users based on a shared starting point, such as their first activity. This analysis helps businesses understand how long users remain active after acquisition and whether retention improves or declines across different cohorts.

When combined, funnel analysis explains *where* users drop off in their behavioral journey, while cohort analysis explains *when* users churn over time. Together, these analyses provide a more comprehensive understanding of customer behavior, allowing businesses to design strategies that address both short-term conversion issues and long-term retention challenges.

LINK TABLEAU :

User Retention Analysis by Monthly Cohort

https://public.tableau.com/app/profile/magasa.faiz/viz/CohortAnalyst_17690837084850/UserRetentionAnalysisbyMonthlyCohort1?publish=yes

https://public.tableau.com/app/profile/magasa.faiz/viz/FunnelAnalysis_17691091824330/TotalUserPerCategory?publish=yes

LINK JUPYTER NOTEBOOK :

https://drive.google.com/file/d/1KsYZ81LA2CMRTCIVUbnkT-qR8h8FaEQX/view?usp=sharing